

SCADA with Remote Monitoring System using GSM Technology

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Abstract—The latest technological trend in industrial monitoring and control is the Supervisory Control and Data Acquisition (SCADA) system. The SCADA systems are the backbone of process automation. To take it a notch further, we integrated a feature that sends data about the power system under SCADA's control to the relevant mobile phone via GSM. Any digital information pertaining to any process automation developed with digital electronics may be automatically sent to any chosen mobile phone anywhere in the globe, thanks to the flexible and modern technology of GSM-based project activities. The primary goal of a supervisory control and data acquisition (SCADA) system is to digitize the analogue data from any power system's many channels, maintain constant watch over all information channels, automatically display the digital data from each channel by selecting the appropriate channel, and safeguard the power system through the implementation of control actions. [4]-[7] Complex devices that need constant vigilance and immediate response from operators greatly benefit from SCADA systems like this. There has been a lot of interest in SCADA systems, and almost every industry is trying to find ways to improve their operations with more sophisticated systems. As a result, we settled on this project's tasks and, to demonstrate our theory in practice, we built a single SCADA system for the power grid. Here, a single power transformer serves as a stand-in for an entire power system. A number of parameters pertaining to this transformer are continuously monitored, converted, and shown digitally. Additionally, control actions are put in place to prevent the transformer from overheating for a variety of reasons. Lastly, all of this data is automatically sent to a chosen mobile phone.

Keywords—SCADA Systems, Power System Automation, GSM-Based Monitoring, Digital Data Acquisition, Real-Time Control

I INTRODUCTION

With the rapid development of technology and the widespread adoption of modern control methods, traditional on-site monitoring approaches are increasingly being replaced by remote supervision systems, which offer numerous operational advantages. Consequently, integrating advanced methodologies and information technology into industrial processes has become essential [1]-[2]. In this paper, the authors present a device designed for remotely transmitting cathodic protection data from oil and gas pipelines to a SCADA system using GSM communication. This device was designed and tested for the first time in Iran at the Iranian Oil Pipelines and Telecommunication Company (IOPTC), Lorestan

Province [3]-[4].

This paper proposes a low-cost, open-source IoT-based SCADA system for monitoring Hybrid Power Systems (HPS) in remote natural gas pipeline control stations. The system replaces expensive, proprietary SCADA architectures with a configurable design that uses voltage and current sensors, an ESP32 microcontroller as the RTU, and a Blynk cloud server as the MTU for data storage and HMI [5]. Data communication is enabled through both GSM (SIM800L) and Wi-Fi configurations to accommodate remote areas lacking stable 3G/4G/Wi-Fi networks. The developed system consumes only 3.9 W and costs 40.1 CAD, demonstrating a highly efficient and economical solution for pipeline monitoring applications [6].

This paper introduces a next-generation, portable IoT-based SCADA system enhanced with GSM connectivity for improved flexibility and reliability. The system is implemented in an industrial setup involving three control processes: temperature, liquid level, and motor operation. Data from these processes is transmitted through a gateway to field supervisors, main supervisors, and monitoring engineers via an IoT cloud platform [7]-[8]. A GSM module provides backup communication during internet outages, and an added camera allows authorized personnel to visually monitor any field location. Practical implementation and testing show that the proposed system performs effectively and meets all operational requirements.

II METHODOLOGY

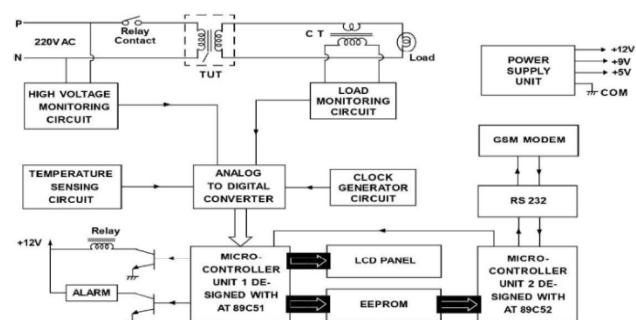


Figure.1 Block diagram of SCADA System

In Fig 1 Utilizing two microcontroller units with varying memory ratings, this project's main controller built with an ATMEL 89C51 chip is responsible for implementing full SCADA operation over the power transformer. It does this by acquiring analog data of the transformer's many parameters and converting it to digital form using an ADC. The data is presented via an LCD screen, which is also interfaced with the same controller through its output port. The ADC is connected to the controller through its input port. The 4Kb of RAM on the controller utilized for this purpose is more than adequate to handle all of the tasks listed above [9]-[10]. The 89C52 is used in the design of the second microcontroller unit, which reads data from the first unit via EEPROM. This chip is designed to transmit all the necessary information about the power transformer to the relevant mobile device via a GSM module; it has 8Kb of memory power. Embedded C, the programming language used to create the code for this device, is more memory intensive than machine code. Through RS232, the chip's output is sent to the GSM modem; the modem accepts TTL logic, which is converted from CMOS logic. Analog electronics are used to track the fundamental signal processing of temperature, load current, and input high voltage parameters associated with distribution transformers. Various analog circuits are designed to transform the output of the sensors into a linear variation that varies in response to the power system condition [11]. The data obtained from the ADC is read by the microcontroller and shown on the screen.

Featuring a 1:1 ratio single-phase transformer, the prototype module's secondary is designed to provide a maximum current of 0.6 amps at 220V AC. The load current may be seen and shown using a current transformer that is linked in series with the load at the secondary. In order to simulate a fault, we also placed an additional load on the demo module. When the load current reached 0.6 amps, the microcontroller cut power to the load to prevent the transformer from overheating. We can see this fault condition on the LCD and send the data to the relevant mobile phone via GSM modem. Equally important are the constant monitoring of the other two parameters, the display of parameter values, the protection of the transformer, and the transmission of fault conditions.

By passing the input voltage to the primary of the transformer via the autotransformer, we can monitor the over voltage parameter. The primary winding of a single-phase transformer can easily burn out at voltages higher than 240V AC, which is why all of these transformers have a relay contact that cuts power to the windings when the voltage gets too high. Typically, the primary of a single-phase transformer is designed to operate at 220V AC, but this isn't always the case. The same holds true for the other two parameters; when their respective limitations are exceeded, the microcontroller's high logic signal activates the relay, cutting power to the main and protecting the transformer from overheating. It is also possible to continually monitor and show the temperature of the transformer's windings or core, which prevents the transformer from overheating and subsequent fire. As a result, the substation environment is replicated in this project work. If a transformer passes any of these criteria, the input to the transformer is removed to prevent it from being burned. Minimizing transformer failures and saving a lot of money is possible with this kind of "SCADA" system installed everywhere at the distribution transformer

end. can use the same approach to manage the power transformer from cell phone with a few tweaks.

One benefit of using a GSM modem is the ability to notify the concerned party via their mobile phone of the transformer's condition or the reason for any failures. Regardless of their location, the concerned party may get the information and respond accordingly by arriving in person or delegating tasks to an underling. This may be put into action by connecting a GSM modem to a pre-programmed microcontroller, which can then transmit the data to a reference mobile phone by SMS. In addition to sending failure information automatically if a defect occurs, up-to-date information on the parameter values may be provided to the cell at regular intervals. Chapter 4 provides a comprehensive explanation of the GSM modem.

Knowing the fundamentals of ATMEL 8-bit microcontrollers is crucial as they are the building blocks of this project. We are now inhabitants of the Embedded World, where a dizzying array of embedded goods made with a wide range of microcontroller chips from various manufacturers ensure that no electrical or electronic equipment can work without one. The smooth operation of these devices is crucial to our everyday lives. The card readers, access controllers, and palm devices at our office make it easy for us to do a lot of what we need to do, much like the television, radio, CD player, washing machine, microwave oven, and many other household appliances. On top of all that, our automobile is equipped with several controls that handle various functions, allowing it to be entirely automated. These controllers are very important in the industrial sector, especially in the areas of SCADA systems and instrumentation. Additionally, the GSM modem requires a microcontroller in order to operate. Many cutting-edge ideas nowadays are built on the idea of integrating GSM with microcontrollers. The ATMEL controller that was utilized for this project is described in depth.

One 8-bit controller that shares architecture with the 8031 core is the ATMEL 89C51. The 8031 architecture from Intel is the most utilized and popular. Many semiconductor manufacturers have developed new products based on this design due to the market popularity of this family. The 8031 is available in a wide number of configurations, and despite its 25-year history, semiconductor companies continue to release products based on it. Data gathering is the first step since the project is centered on three critical and potentially dangerous power transformer characteristics that the system is meant to measure. We will not be taking into account the other drifting parameters—efficiency, frequency, power factor, etc.—in this project since they pose no threat to the transformer. Three separate transducers are used to separately collect data from the three channels:

A current transformer (CT) could be used to measure the load current; this type of transformer has its primary connected in series with the load, and its secondary generates a proportionate AC voltage that is converted to DC through a current adjusting pot. The voltage is then fed into an A/D converter. A temperature sensor that measures the transformer's body temperature uses a semiconducting device; as the temperature increases, the device's resistance decreases, and the difference in resistance is amplified by an operational amplifier set up as a differential amplifier. There is a linear relationship

between the temperature and the ultimate output of this circuit.

A potential transformer (PT) is used to measure the primary voltage of the transformer; this device displays the calibrated fluctuation in the secondary voltage (which is affected by changes in the line voltage).

Parameter values determine the DC voltages that are generated from the sensor outputs. For instance, the output voltage increases extremely linearly with the increase in transformer body temperature because the circuit of each channel is intended to achieve high linearity. Basically, the microcontroller can't handle analog data, thus it has to be converted to digital. The next step, then, is to digitize the analogue data collected from the outputs of each channel. This is accomplished by using an 8-channel ADC 0809 IC. The time required for the conversion is directly proportional to the frequency that the timer IC generates, and it is a very responsive IC.

According to the first chapter, the project is built around two microcontroller units. These units are functionally identical, with the exception of their memory power. 89C51 and 89C52 are 8-bit CMOS controllers with flash programmable and erasable read-only memory chips; the former is used in the SCADA system, while the latter is used for data transfer using GSM modem. With 40 pins and 32 I/O lines apiece, these devices are ideal for interfacing with a wide variety of electronic gear. There are a lot of things that the 89C51-designed microcontroller unit 1 can do. Listed below are the many functions.

Acquiring digital data from the A/D converter is the primary purpose of this microcontroller device.

Using an LCD display to show the value of the parameter. Showing the reason for each parameter's failure on the same LCD. Preventing damages to the power transformer from overvoltage, overheating, and overload. This means that the controller will trigger the relay and cut power to the transformer the second any parameter value goes over the threshold. Data is saved in an EEPROM as soon as it becomes accessible. The second microcontroller then accesses this data and sends it to the relevant mobile phone via GSM modem.

The system is now fully functional as a SCADA system, thanks to the first microcontroller performing these operations. Each parameter has a pre-programmed value, and the controller will instantly activate the relay and cut power to the transformer main if any parameter value is greater than the preset value. If the power system's transformer has a rating of 600 milliamps (0.6 Amps), then the load applied to the secondary should not exceed 132 Watts ($0.6 \times 220 \text{ V}$). If the load does exceed this rating, the microcontroller will activate the relay, display the information on the

LCD, and send the same signal to the mobile device. The main supply to the transformer may be cut off via the relay contact, preventing the transformer from overheating and perhaps burning. The other two parameters are also continually monitored; similarly, the line voltage is set at 240 V AC and the temperature at 50 °C. When a parameter value goes over its preset value, an alarm is immediately activated at the microcontroller's output to notify the repair crew.

III RESULTS AND DISCUSSION

A current transformer is used for continuous monitoring of

the load current. In order to measure current, one uses a current transformer. The CT secondary produces a voltage that is proportional to the current flowing through the primary winding, which is connected in series with the load. Due to the small number of turns in the main winding, the voltage loss across it is negligible. The number of turns in the secondary winding of a current transformer is bigger; the ratio of the turns determines the precise number. For this calibration, the ratio is 1:50, which generates a respectable voltage at the CT secondary. After being rectified and filtered, the CT's output is sent to an A/D converter, which transforms the analog data of the current passing through the CT's main into digital data. From this data, the microcontroller calculates the current through CT primary and displays it in amps. The microcontroller is set up such that it automatically cuts power to the transformer primary if the load on the secondary exceeds the 0.6 amp rating, which is the maximum current that the power transformer is rated to deliver. Consequently, the transformer is shielded against overload- related fires.

A. High Voltage Monitoring Circuit

Overvoltage is one of the several potential reasons for a transformer to fail. If the voltage applied to the primary windings of a distribution transformer—or any other kind of transformer—is higher than its rated voltage, the transformer might quickly catch fire from overvoltage. In this project, a circuit for monitoring and controlling voltage is used to prevent the transformer from overheating and burning.

In this project, we utilize an autotransformer to generate high voltage since we can't control the line voltage; the idea is to prevent the power transformer from overheating and burning. As a line voltage sensor, a 6V- 0-6V center-tapped secondary step-down transformer is used. An analog-to-digital converter takes the rectified and filtered secondary voltage, which rises in tandem with the main voltage of the transformer, and uses it to translate the analogue data into its digital equivalent.

B. Temperature Sensing Circuit

There are primarily two types of temperature measuring techniques, based on whether the body being tested and the hot system exchange heat by direct touch or through radiation sent over a medium. Thermocouples and other contact techniques rely on physical touch to create a thermodynamic equilibrium between a heated body and a testing body. This is used in thermometers and other devices submerged in solids or liquids. Radiation released when excited atoms and molecules in the heated body return to the ground state establishes the thermodynamic equilibrium in non-contact techniques (used, for example, in radiation pyrometers).

For temperature measurement, there are a variety of tools and techniques that rely on either touch or non- contact techniques. Also, there are many different kinds of sensors on the market; picking the right one depends on the instrument using. A device of the RTD (Resistance temperature detector) type is used for the purpose of monitoring temperature in this project. The 'SL100' NPN general purpose switching transistor serves this function. A transducer of this kind may transform thermal energy into a form with a changeable resistance. The RTDs work by using the idea that a material's electrical resistance would vary in direct correlation to the applied temperature change. The resistance of this RTD's metal detecting

element grows in direct proportion to the temperature. These RTDs are ideal for surface temperature monitoring because of their many benefits, such as their physical and chemical durability, acute sensitivity at ambient and lower temperatures, and their ease of mounting Fig 2.

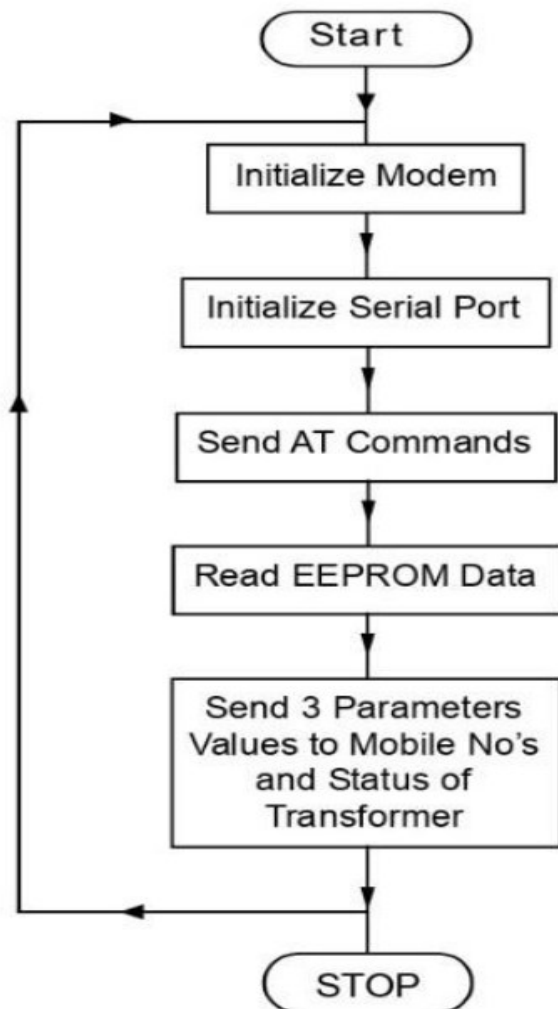


Figure.2 Flow chart of SCADA System

An ATMEEL 89C52 microcontroller chip forms the basis of the microcontroller unit. 8K bytes of flash programmable and erasable read-only memory (PEROM) makes this low-power, high-performance CMOS 8-bit microcomputer tick. As a robust controller with a high-density non-volatile memory that is compatible with the standard MCS-51 instruction set, it offers a cost-effective and highly versatile option for control applications.

The microcontroller executes the instructions given to it by the programme. The Harvard design laid out the four pillars of an embedded system and is the basis for the majority of modern micro controllers. Integrated circuits typically consist of a central processing unit (CPU), read-only memory (ROM or Flash memory), random-access memory (RAM), a timer (or timers), and input/output (I/O) connections for connecting to external peripherals and supplementary resources. The main difference between a general-purpose CPU chip and a microcontroller is that the former often requires less peripheral support chips to function as a computer. Simply inserting the microcontroller into the device to be controlled, connecting it to power and any necessary data, and that's all there is to it.

The Global System for Mobile Communications, or GSM,

is a network that allows for digital cellular phone calls. While its original intent was to provide a uniform mobile phone standard throughout

Europe, its widespread adoption has since surprised even its creators Fig 3. In many instances, GSM is light years ahead of the previous public switched telephone network and offers consumers a full suite of services and capabilities that were previously unavailable on analogue cellular networks. Advanced services and features, such as global roaming on other GSM networks, are also a part of GSM, in addition to digital transmission Fig 4.

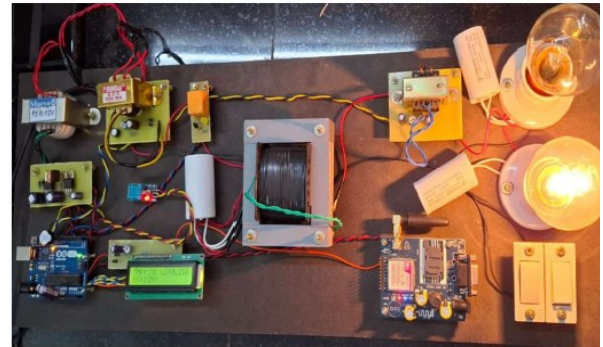


Fig 3 Prototype of SCADA with remote monitoring system using GSM technology kit.

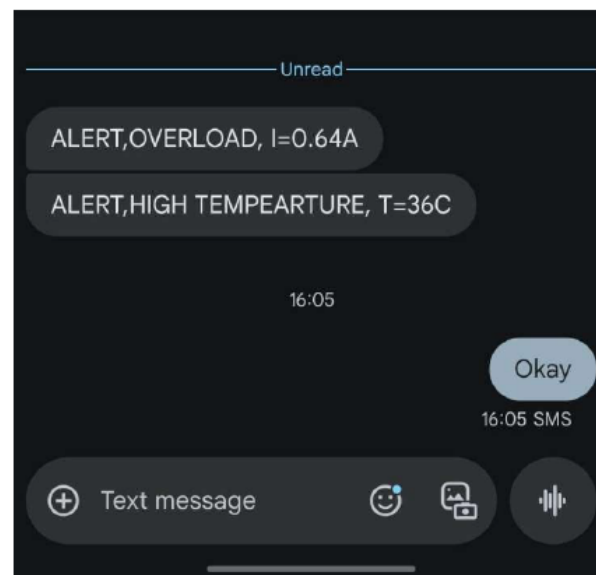


Figure.4 SMS Alert Functionality In Scada System

IV CONCLUSION

Improving SCADA system technology by sending and showing limitless data to the relevant mobile phone via GSM modem is the primary goal of this project. In this notion, a single microcontroller chip may accomplish all the work of a computer, including monitoring, regulating, processing, storing data, and presenting massive amounts of information. Therefore, no computer is needed for this project; the microcontroller serves as a minicomputer and does all of the aforementioned tasks. To improve the technology, a methodical, sequential approach is used. A single GSM modem and two microcontroller units comprise the whole SCADA system. Given the practical nature of our study, we have built a prototype module

specifically for the live presentation. The findings are good, and this module is pretty close to the genuine operating system. In order to transmit data to the mobile phone, the second microcontroller unit—which is responsible for reading from the EEPROM—is meant to talk to the GSM modem via RS232. In this case, data may be transmitted to several phones at once; hence, the system has to keep the matching phone numbers. The controller reads data from EEPROM and displays it according to the power system circumstances. Its design prioritizes minimalism in hardware compared to current systems. Implementing simple technology that gives numerous functionalities is a good idea. Any power system may be used to construct this system with minor adjustments. Because EEPROM data is persistent and will not be erased in the event of a power outage, the controller may always access this data anytime the system is powered on. The system may now be considered innovative thanks to the aforementioned changes.

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